

Usama Bin Ejaz

AI/ML Engineer | LLMs, RAG, AI Agents | Data Scientist

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SUMMARY

AI/ML Engineer with **4+ years** of experience architecting production-grade machine learning systems, LLM applications, and scalable data infrastructure across e-commerce, fintech, logistics, HR/SaaS, and property valuation domains. Proven track record of measurable business impact: **85% LLM extraction accuracy**, **18% fraud reduction**, **20% forecasting improvement**, and **30% customer engagement lift**. End-to-end ownership from data pipeline architecture through model deployment to full-stack application delivery. MS in AI & Data Science (Germany), fluent in English (IELTS C1).

TECHNICAL SKILLS

AI / ML

TensorFlow PyTorch Scikit-learn Hugging Face OpenAI API LangGraph LangChain Pinecone FAISS
RAG NLP Computer Vision Prompt Engineering AI-Assisted Engineering (Claude Code, Cursor)

Data Engineering

Airflow PySpark dbt ETL/ELT BigQuery Redshift AWS Lambda Prophet ARIMA

Languages & Cloud

Python SQL TypeScript JavaScript C++ AWS GCP Azure PostgreSQL MongoDB MySQL

Web & Visualization

FastAPI Next.js React React Native Electron Prisma Tailwind CSS Power BI Tableau Recharts

PROFESSIONAL EXPERIENCE

Data Scientist

Jun 2026 – Present

AiGenix • Karachi, Pakistan

- Own the **data value chain** of PerkMap, a geospatial perks-discovery platform: built a crawling + **LLM-extraction pipeline** converting unstructured promo pages into a deduplicated (fuzzy-matched), geocoded deals catalogue with per-day cost gating
- Designed the intelligent features: a **location-aware GPT discovery assistant** (structured JSON outputs + web search) and multimodal **poster-OCR auto-fill**, behind a swappable multi-provider LLM abstraction
- Built **merchant analytics**: audience dashboards and a privacy-preserving geospatial activity heatmap with consent gating and **small-cohort suppression**
- Shipped the delivery surfaces end-to-end: a **5-city** consumer app (Expo/React Native, **live on Google Play**; iOS in review) and a bilingual (EN/AR) merchant portal on a Supabase/Postgres API with JWT + Row-Level Security

Freelance AI/ML Engineer

2022 – Present

Independent & collaborative client projects • Remote

- Deliver **computer-vision and LLM systems** for international clients across conservation, infrastructure, logistics, and healthcare
- Highlights: endangered-species detection (**~97% mAP@50**), geospatial road-damage detection for a Dutch municipal road authority, license-plate access control for an Amsterdam logistics warehouse, and a biomarker lab-report insights API

Data Scientist

Oct 2025 – Apr 2026

Dellsons Associates • Karachi, Pakistan

- Led AI/ML delivery on a compressed product roadmap — shipped **4 production systems in 7 months** across enterprise products
- Engineered an **AI-powered email command center** (V3, 5-account multi-mailbox) with LangGraph agent orchestration, local Mistral 7B summarization, importance scoring engine, relationship intelligence, and automation rules — **reducing manual email processing time by ~60%** for a 50-person organization
- Architected a **cross-platform team collaboration suite** (web + mobile) with real-time chat, RBAC, and push notifications, **consolidating 3 separate tools into one platform** for 30+ daily active users
- Built a **RAG-powered document Q&A system** with Pinecone vector search and source attribution, enabling instant retrieval across 10K+ documents — **cutting research time from hours to minutes**
- Delivered a **BI & compliance dashboard** with dynamic scoring algorithms and automated Excel reporting, **eliminating 20+ hours/month** of manual compliance tracking for the property valuation industry

AI/ML Engineer

Mar 2022 – Sep 2025

Zapdas Technologies • Karachi, Pakistan

- Owned full ML lifecycle across two SaaS products (e-commerce platform & staff management system), from model development and pipeline architecture to production deployment and monitoring
- Deployed **time-series forecasting** (ARIMA, Prophet), **anomaly detection**, and **recommendation systems** — delivering **20% forecasting accuracy improvement**, **18% fraud reduction**, and **30% customer engagement lift**

- Pioneered **LLM-powered invoice extraction** using Llama 3.2 Vision, achieving **85% accuracy (30% above traditional OCR)**, automating data entry for 500+ monthly invoices
- Architected **production data infrastructure** — metadata-driven ingestion pipelines into BigQuery, multi-source data lake on AWS (Redshift, Lambda), and automated ETL workflows **reducing manual workload by 20%**
- Developed AI chatbots (OpenAI) that **improved customer resolution time by 25%**, and built predictive HR models for attrition and geospatial workforce optimization **cutting operational inefficiencies by 15%**

Data Analyst (Work Student)

Mar 2024 – Aug 2024

TDK Electronics • Munich, Germany

- Developed Prophet-based forecasting models for electronic component failure prediction, directly improving inventory planning and reducing supply chain stockouts
- Designed interactive analytical dashboards (Excel + Tableau) and delivered data-driven insights to senior stakeholders and external clients

Data Analyst Intern

Sep 2021 – Feb 2022

Swvl • Karachi, Pakistan

- Built demand forecasting and geospatial models for fleet optimization — **reduced idle time by 10%** and **improved ride availability by 14%**
- Developed predictive complaint classification system that **cut escalation rates by 24%**
- Automated cross-platform reporting (Google Sheets, BigQuery, Power BI, Metabase) — **boosted operational efficiency by 15%**

KEY PROJECTS

NeuroMail V3 — AI Email Command Center

Shipped Product • Dellsons

Electron • Python • FastAPI • React • LangGraph • Mistral 7B • OpenAI • Gmail/Outlook SDK • SQLite

Architected a unified multi-mailbox desktop app (up to 5 accounts, Gmail + Outlook) with **LangGraph multi-agent orchestration** for autonomous triage and draft generation. Built an **AI morning briefing** with composite importance scoring (sender/content/metadata/recency), **relationship intelligence** panel, cross-account **automation rules engine**, email analytics dashboard, **local Mistral 7B thread summarization**, smart unsubscribe manager, snooze/reminders, and Ctrl+K command palette.

PerkMap — Card-Perks Discovery Platform

Live Product • AiGenix

LLM Extraction (Groq/OpenAI) • Supabase • PostgreSQL • Expo/React Native • Next.js

Geospatial perks-discovery platform, DS-led: **crawler + LLM-extraction pipeline** building a deduplicated, geocoded deals catalogue; **location-aware GPT discovery assistant** with structured outputs; privacy-preserving merchant analytics heatmap with small-cohort suppression. **Live on Google Play across 5 cities**; iOS in review.

RAG-Powered Document Q&A System

Production • Dellsons

Python • Pinecone • FAISS • OpenAI • MySQL • FastAPI

Built an end-to-end retrieval-augmented generation platform enabling **conversational Q&A** over large document repositories.

Engineered a **vector embedding pipeline with Pinecone** for semantic search, implemented threaded chat with full source attribution and chunk tracking, and designed multi-tenant namespace isolation with content hashing for deduplication.

E-commerce Analytics & ML Platform

Production • Zapdas

Python • BigQuery • AWS • Redshift • FastAPI • TensorFlow • Prophet • OpenAI

Developed a comprehensive ML and analytics layer for an e-commerce platform encompassing **time-series demand forecasting**, **real-time anomaly detection**, **LLM-powered invoice extraction** (Llama 3.2 Vision), personalized recommendation engine, conversational AI chatbot, and a **metadata-driven ETL pipeline** feeding clickstream and transactional data into BigQuery.

EDUCATION

MS in Artificial Intelligence & Data Science

Mar 2023 – Aug 2024

Deggendorf Institute of Technology — Deggendorf, Germany

BS in Computer Science

Aug 2018 – Jun 2022

FAST-NUCES (National University of Computer & Emerging Sciences) — Karachi, Pakistan

CERTIFICATIONS & LANGUAGES

IELTS C1 Advanced • **SQL** Basic to Advanced (HackerRank, 2025) • **Supervised Learning** (DataCamp, 2021) • **Data Science** Next Gen Analytics (2022) • **Data Visualization in Python** (DataCamp, 2020)

Languages: English (Fluent) • Urdu (Native or Bilingual)